

Block-encoding K matrices A_k of size $2^n \times 2^n$

$K = 8$ blocks; n sweep = [2, 3, 4, 5, 6, 7]; phase_bitsize $b = 32$.

Approach 1 (SVD-based):

$A_k = U_k \Sigma_k V_k$. Block-encoded as

$\begin{bmatrix} \text{sum}_k |k\rangle\langle k| (x) U_k \\ \text{diag}(\text{sigma}) + \text{ancilla rotation} \end{bmatrix} \cdot \begin{bmatrix} \text{sum}_k |k\rangle\langle k| (x) V_k \end{bmatrix}$.

- Interferometer cost for U_k and V_k :

BlockUnitaryInterferometerSynthesisQROAM($n_{\text{blocks}}=K$, $n_{\text{rows}}=2^n$, $n_{\text{layers}}=2^n$).

- Diagonal cost: QROAMClean($\text{data_shape}=(K, 2^n)$, b) + ctrl-AddIntoPhaseGrad(b, b) + QROAMCleanAdjoint.

- Toffoli total = $2 \times \text{interferometer} + \text{diagonal}$.

- Qubit total $\sim \max(\text{interferometer}, \text{diagonal}) + 1$ (block-encoding ancilla).

Approach 2 (direct dilation):

Each A_k is completed (Stinespring) to a unitary W_k of size $2^{(n+1)} \times 2^{(n+1)}$.

Synthesize $\text{sum}_k |k\rangle\langle k| (x) W_k$ via one interferometer on $(n+1)$ system qubits:

BlockUnitaryInterferometerSynthesisQROAM($n_{\text{blocks}}=K$, $n_{\text{rows}}=2^{(n+1)}$, $n_{\text{layers}}=2^{(n+1)}$).

Subnormalization $\alpha = 1$.

Approach 3 (reflection rectangular):

$B = (H_A (x) I) \cdot \text{SWAP}_{\{A, \text{col}\}} \cdot O_A \cdot (H_A (x) I)$ where

$O_A = \text{QROAMClean}((K, N, N), b) + \text{ctrl-AddIntoPhaseGrad} + \text{QROAMCleanAdjoint}$

loads the entry-rotation angle. Subnormalization $\alpha = N = 2^n$: the raw Toffoli is small but algorithms (QSVT, QPE) need an extra factor N .

All resource counts come from Qualtran's `get_cost_value(QECGatesCost())`, via `utils.get_Toffoli_counts` and `utils.get_qubit_counts`. Each cost is optimized over `log_block_size` in `0..11`.

Two optimization modes per approach are shown: Toffoli-optimal and qubit-optimal.

Fits $y = c \cdot 2^{(\alpha \cdot n)}$:

SVD, Toffoli-opt T: $\alpha=1.219$, $c=3.207e+02$; Q: $\alpha=0.384$, $c=3.422e+01$

SVD, qubit-opt T: $\alpha=1.311$, $c=2.633e+02$; Q: $\alpha=0.038$, $c=6.743e+01$

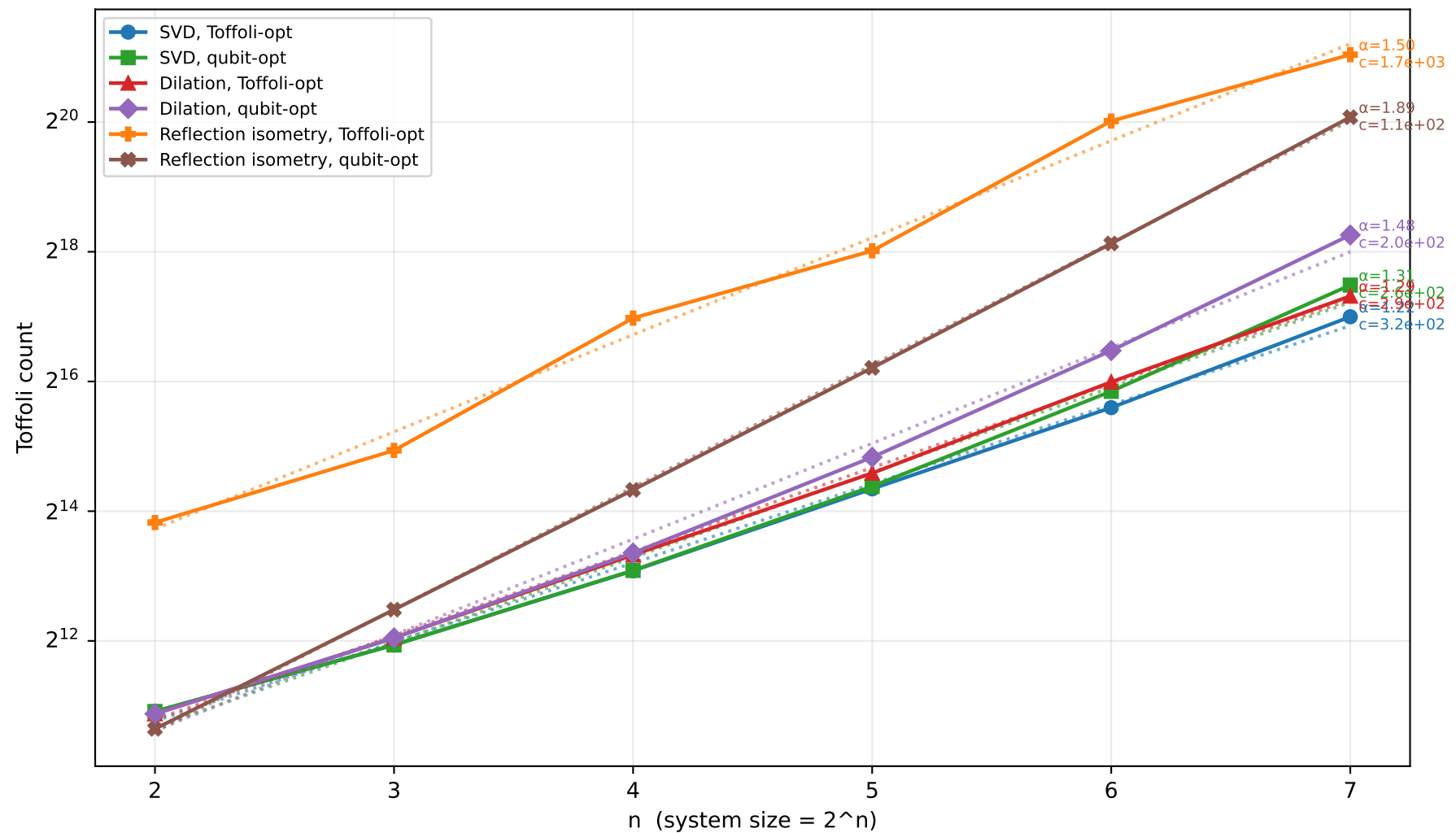
Dilation, Toffoli-opt T: $\alpha=1.294$, $c=2.939e+02$; Q: $\alpha=0.431$, $c=3.688e+01$

Dilation, qubit-opt T: $\alpha=1.477$, $c=2.027e+02$; Q: $\alpha=0.037$, $c=6.942e+01$

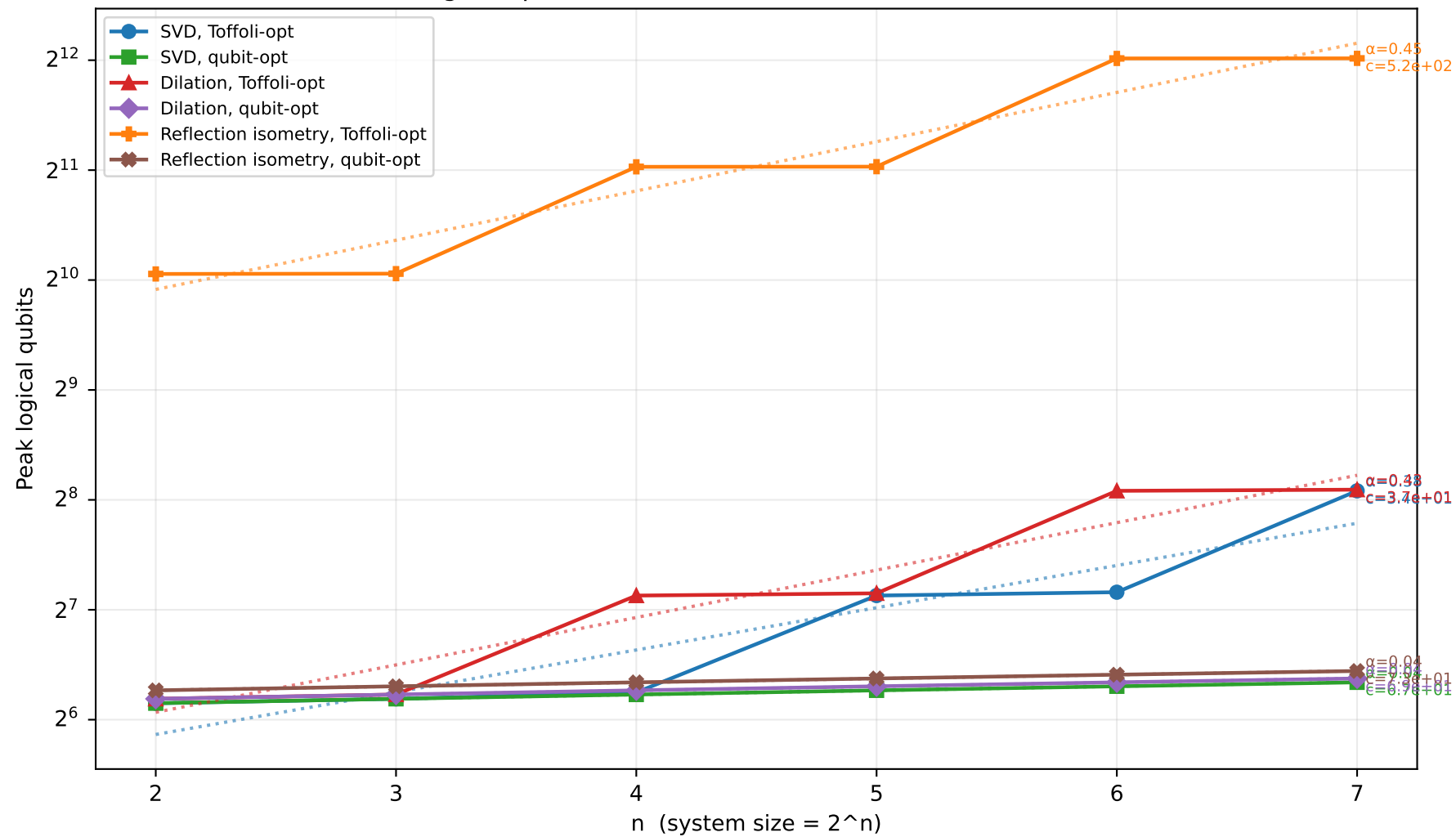
Reflection isometry, Toffoli-opt T: $\alpha=1.495$, $c=1.709e+03$; Q: $\alpha=0.448$, $c=5.185e+02$

Reflection isometry, qubit-opt T: $\alpha=1.885$, $c=1.137e+02$; Q: $\alpha=0.035$, $c=7.340e+01$

Toffoli count: SVD vs direct dilation (K=8 blocks, b=32)



Peak logical qubits: SVD vs direct dilation (K=8 blocks, b=32)



Resource counts (K=8, b=32)

n	dim	SVD T-ops	SVD T-ops	SVD Q-ops	SVD Q-ops	Dil T-ops	Dil T-ops	Dil Q-ops	Dil Q-ops	Refl T-ops	Refl T-ops	Refl Q-ops	Refl Q-ops	alpha_ref
2	4	1,926	71	1,926	71	1,879	73	1,879	73	14,520	1064	1,600	77	4
3	8	3,922	73	3,922	73	4,231	75	4,231	75	31,368	1066	5,720	79	8
4	16	8,666	76	8,698	75	10,264	140	10,487	77	128,992	2091	20,576	81	16
5	32	20,804	140	21,346	77	24,600	142	29,175	79	264,800	2093	75,808	83	32
6	64	49,556	143	58,994	79	65,178	271	91,191	81	1,060,096	4142	286,464	85	64
7	128	130,856	271	183,554	81	163,226	273	313,655	83	2,155,136	4144	1,104,768	87	128